

S. SHERIFRED SINGH

Embedded Software Engineer

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PROFESSIONAL SUMMARY

- Embedded Software Engineer with **6.5+ years of total industry experience**, including **4.5+ years dedicated to Embedded C firmware development** and 2 years in hardware testing, board bring-up, and PCB fault diagnostics.
- Experienced in developing **bare-metal firmware** and **low-level device drivers** for **PIC, ARM Cortex-M (STM32), and ESP32** microcontrollers across real-time industrial automation and embedded control systems.
- Proficient in peripheral interfacing and hardware communication protocols including **UART, SPI, I2C, RS-232, ADC, Timers, PWM, GPIO, External Interrupts, and Non-Volatile Memory (EEPROM)**.
- Strong foundation in embedded architecture, interrupt-driven firmware design, **Finite State Machine (FSM)** implementation, **register-level programming**, and SDLC best practices.
- Skilled in **hardware-software integration, board bring-up, signal verification**, and resolving component-level PCB faults using **digital storage oscilloscopes, logic analyzers, and multimeters**.

TECHNICAL SKILLS

- **Microcontrollers** : ARM Cortex-M3 (STM32F103), Microchip PIC (18F46K22, 16F887), ESP32
- **Programming Languages** : Embedded C, C
- **Communication Protocols** : UART, SPI, I2C, RS-232, RS-485
- **MCU Peripherals** : ADC, Timers & Counters, GPIO, Interrupts (ISR), PWM, EEPROM, Flash Memory
- **Embedded Development** : Bare-Metal Firmware, Low-Level Driver Development, Hardware-Software Integration, State Machine Design (FSM), Real-Time Debugging
- **Testing & Validation** : Firmware Verification, Integration Testing, Hardware-in-the-Loop (HIL) Validation, Fault Isolation
- **Embedded Concepts** : Interrupt Handling & Latency, Embedded Architecture, Memory-Mapped I/O, Watchdog Timers (WDT), SDLC Fundamentals
- **Sensors & Drivers** : RTD PT100 (MAX31865 SPI), Thermocouples (MAX6675 SPI), NTC 10K, GSM/GPRS (SIM800A), RTC (DS3231/DS1307)
- **Development Tools** : MPLAB X IDE, STM32CubeIDE, Keil uVision, Git Version Control
- **Hardware Skills** : Board Bring-Up, PCB Testing & Debugging, Soldering (SMD & Through-Hole), Oscilloscope (DSO), Multimeter, Power Management

PROFESSIONAL EXPERIENCE

Embedded Software Developer

Jan 2022 – Present

NK Automation, Coimbatore

- Architected and maintained **bare-metal Embedded C firmware** for **PIC18 and STM32** microcontrollers used in industrial automation and real-time control systems.
- Designed and implemented modular, interrupt-driven **low-level peripheral drivers** for **UART, SPI, I2C, ADC, Timers, PWM, EEPROM**, and display controllers (**LCD, GLCD**).
- Integrated industrial sensors and external memory subsystems including **PT100 RTDs, thermocouples, DS3231 RTC**, and **I2C EEPROM** for reliable data acquisition and event logging.
- Developed deterministic **Finite State Machine (FSM) control algorithms** for industrial production monitoring, closed-loop temperature control, and **Automatic Mains Failure (AMF) & Generator Control** systems.
- Optimized firmware for improved execution speed, memory utilization, and system reliability through efficient **register-level programming** and **non-blocking software design**.
- Performed **hardware bring-up, hardware-software integration, signal verification**, and system validation using **digital storage oscilloscopes, multimeters, and serial bus analyzers**.
- Collaborated with cross-functional teams to troubleshoot field issues, analyze failure logs, validate embedded systems, and deliver reliable industrial automation solutions.

Testing Engineer

Jun 2018 – Jun 2020

Akarsh Technologies, Coimbatore

- Performed precision **soldering and desoldering on SMD and through-hole PCBs**, supporting hardware validation, **PCB debugging, fault isolation**, and component-level rework.

- Analyzed and resolved complex PCB faults, **power rail short circuits**, and component failures in textile and industrial automation systems.
- Conducted **hardware-in-the-loop firmware validation** and hardware functionality testing to verify design specifications and eliminate defects prior to production release.

TECHNICAL PROJECTS

Advanced Multi-Channel Powerloom Monitoring System—*PIC18 MCU, Embedded C, I2C, EEPROM, RTC, GLCD*

- Architected an **interrupt-driven embedded firmware system** for real-time industrial textile production tracking, capturing high-speed machine events with **zero data loss**.
- Developed **dynamic calculation algorithms** for critical production metrics (picks, meters, beam balance, and operator pay) with a responsive **GLCD-based menu navigation interface**.
- Integrated **DS3231 RTC** and **AT24C32 EEPROM via I2C** protocol for shift-wise data logging, ensuring **100% seamless data retention and recovery** during industrial power failures.
- Executed end-to-end **hardware-software integration**, **noise immunity validation**, and rigorous field testing in high-noise industrial environments.

Digital Temperature Controller using PT100 + LCD Display—*PIC MCU, MAX31865, SPI, Embedded C*

- Developed Embedded C firmware interfacing **PT100 RTD via MAX31865 SPI driver** for high-precision temperature acquisition.
- Implemented a modular LCD driver for real-time visualization of temperature, setpoint, and system power status.
- Designed a **closed-loop relay control algorithm with software hysteresis** to regulate heater operation and eliminate contact chatter.
- Optimized system using **timer interrupts for stable ADC sampling**, switch debounce handling, and flicker-free LCD refresh routines.

Intelligent Auto Mains Failure (AMF) & Generator Controller —*PIC MCU, Embedded C, SPI, Timers, Relays*

- Architected interrupt-driven firmware achieving **~98% reliable detection** of grid undervoltage and phase imbalance using **calibrated ADC sampling with noise-filtered voltage sensing**.
- Implemented a **deterministic multi-state control algorithm** for reliable grid-to-generator switching with safe **break-before-make interlocking logic**.
- Optimized generator control routines (cranking, re-crank, warm-up, cool-down) with **EEPROM-based configuration storage**, reducing false crank attempts by **~40%**.
- Engineered **fail-safe protection layer** with relay control, phase validation, and fault-handling logic, reducing unsafe switching by **over 60%**.

CERTIFICATIONS & TRAINING

Diploma in Embedded Systems

Jun 2020 – Jan 2021

Manfree Technologies, Coimbatore

- Hands-on practical training in **ARM Cortex-M (STM32F103C6)**, **ESP32**, and **PIC** microcontroller programming using Embedded C and C++.
- Peripheral interfacing: **DS1307 RTC**, **RS-232**, **GSM (SIM800A)**, **UART**, **SPI**, **I2C**, and Bluetooth communication modules.
- Tools and toolchains: **Keil uVision**, **MPLAB X IDE**, **DevC++**, and **STM32CubeIDE**.
- Completed applied mini-projects in analog sensor acquisition, communication modules, and real-time embedded control systems.

EDUCATION

Coimbatore Institute of Technology

2021 – 2024

B.E. in Electrical & Electronics Engineering (Part-Time) | Coimbatore, India | 72%

(Completed degree through Evening/Part-Time Program concurrently with full-time professional engineering employment)

Government Polytechnic College, Anna University

2015 – 2018

Diploma in Electronics & Communication Engineering | Coimbatore, India | 72%

Vimal Jyothi Convent Matric Higher Secondary School

2015

Higher Secondary Certificate | Coimbatore, India | 71%